

Prospects of Bagasse Cogeneration in Sugar Industries of Nepal

Sajjan Pokhrel^{*1}, Saroj Bhattarai¹, Srijan Rajbamshi¹ and Yogesh Bhusal¹¹IOE, Pulchowk campus

Abstract - Bagasse cogeneration describes the use of fibrous sugarcane waste, bagasse, to cogenerate heat and electricity at high efficiency in sugar mills. This report indicates that there is abundant opportunity for the wider use of bagasse-based cogeneration in sugar industries of Nepal and this, in addition to making these sugar mills power independent, can contribute electric power substantially to the national grid. Yet this potential remains largely unexploited.

The research evaluates the technical feasibility and economic viability of reconfiguring the sugar industries towards cogeneration and also quantifies the emissions from Bagasse cogeneration. This report depicts the Bagasse cogeneration potential of Nepalese sugar industries: the total electric power that can be produced and fed to the national grid, the economic issues and the issues of emissions.

Index Terms - ROR (Run of the river), NEA(Nepal Electricity Authority), TCD (Tones of crushing of canes per day), NCV(Net calorific value), GCV(Gross calorific value).

I. INTRODUCTION

Nepal is being ravaged by the electricity crisis. The electricity crisis of this millennium began in 2006. Nepal saw the last electricity crisis of the last millennium in 1999 and, with the commissioning of Khimti project in 2000, there was no load shedding till 2005. Since then, we have been facing load-shedding of upto 16 hours a day in the dry season, and 300 MW of peak time electric shortage even if the installed plants run at full capacity (Nepal Electricity Authority, 2013) This is because most of the hydropower projects of Nepal are ROR type, which are snow fed, and get dry during the months of dry season. This power supply shortfall may become worse as demand growth exceeds the ability of utilities to finance new capacity. Reduced levels of electric service and reliability are impeding economic expansion and forcing utility customers who can afford them to install costly and high maintenance cost backup generators.

The time has come that, the country must search now for the alternative sources for electricity generation, like renewable and non-conventional sources such as solar, wind, or hybrid technologies.

One potential low-cost source of future supplies is industrial cogeneration, and several studies in India, as well as experience in other parts of the world, point to the sugar industry as a prime candidate for this form of supplemental power production.

It is clear that the sugar industries in Nepal now have to chart a strategic course that will manage the evolution of technology in the sugar power plants in the most beneficial

way. The Government of Nepal and NEA will have to play a very significant role in that process if the country and the utility are to maximize the benefits that would accrue. Bagasse would be used during the crop season and rice husk during the off-crop season. Rice husk usage during the off-season will enable the exportation of extra power to the grid.

Advantages include relatively low capital requirements, a renewable, indigenous waste material as a fuel, and sugar mills of sufficient number and size to make a measurable contribution to power supplies. Due to its low sulphur content, SO_x formation from bagasse was lower than that from fossil-based fuels, and therefore such usages of renewable resources could alleviate CO₂ emissions.

II. COLLECTED DATA

Table 1: Collected data sample from major three sugar industries of Nepal.

S. N	Particulars	Indushanker	Everest	Shreeram
1	Plant capacity (TCD)	3,000	5,000	3,500
2	Cane crushed last year (Metric tons)	263,100	454,784	186,000
3	Operational days last year	128	120	110
4	Actual TCD	2,055	3,790	1,691
5	Plant utilization (%)	68.52%	75.80%	48.31%
6	Sugar recovery (%)	9.00%	9.00%	9.00%
7	Recovered sugar(MT)	23,679	40,931	16,740
8	Molasses generation (%)	5%	4.50%	5%
9	Molasses produced	13,155	20,465	9,300
10	Bagasse generation (%)	34%	34%	34%
11	Bagasse generated	89,454	154,626	63,240
12	Bagasse used (%)	79%	82%	100%
13	Bagasse used	71,037	127,339	63,240
14	Excess bagasse (as % of cane crushed)	7%	6%	0%
15	Excess bagasse	18,417	27,287	0

Table 2: Rice husk production in Terai region.

Terai Region	Paddy (MT)	Husk (MT)
Eastern Terai	1,117,600	223,520
Western Terai	699,687	139,937
Central Terai	1,080,237	216,047
Mid-Western Terai	469,826	93,965
Far Western Terai	347,680	69,536
Total	3,715,030	743,006

Source: (Agribusiness Promotion and Statistics Division , 2011-2012)

* Corresponding author: 111pokhrel@gmail.com

III. DATA ANALYSIS

3.1 Working formula

Power analysis

1. Power calculation

$$P = \dot{m} \times (h_2 - h_1)$$

Where,

$\dot{m}$ = mass flow rate.

$(h_2 - h_1)$ = change of enthalpy

2. Calorific value

- a. Net calorific value

$$NCV = [(18,309 - 2076(\text{moisture \% sample}) - 19605(\text{ash \% sample}) - 31,14(\text{brix \% sample}))] \text{ kJ/kg}$$

- b. Gross calorific value

$$GCV = [(19,605 - 196,05(\text{moisture \% sample}) - 196,05(\text{ash \% sample}) - 31,14(\text{brix \% sample}))] \text{ kJ/kg}$$

3. Evaporation ratio

- a. Gross Evaporation coefficient

$$x = \frac{\eta_B \cdot HCV}{(h_{st} - h_{fw})}$$

Where,

η_B = Efficiency of boiler

HCV= High or Gross calorific value

h_{st} = enthalpy of steam generated

h_{fw} = Enthalpy of feed water steam

- b. Evaporation ratio

$$y = \frac{x \cdot (F\%C - \text{FibreLost}\%C)}{F\%B}$$

Where.

F%C = Percentage of fiber in cane

F%B = Percentage of fiber in bagasse

FL%C = Fiber lost during milling

4. Calculating the quantity of steam generated.
5. Steam generated = Total bagasse $\times$ Evaporation ratio
6. Calculating Mass flow rate
7. Mass flow rate = $\frac{\text{Total steam generated}}{\text{Total hour of operation}}$
8. Calculating the enthalpy of steam using Mollier's chart
9. Calculating the total heat energy of steam at inlet and outlet
10. Total heat energy at turbine inlet = Mass flow rate $\times$ Enthalpy of steam

11. Total heat energy of back pressure steam = Mass flow rate $\times$ Enthalpy of steam

12. Calculating heat energy available for power generation

13. Surplus heat energy = Energy of steam at turbine inlet – Back pressure steam heat energy

14. Calculation of power generation

As we know that relation between kcal and kWh

i.e. 860 kcal = 1kWh

15. Total power generation = $\frac{\text{surplus heat energy}}{860} \times \eta_t \times \eta_a \times \eta_{tp}$

Where,

η_t = Thermal efficiency of turbine

η_a = Alternator efficiency

η_{tp} = Power transmission efficiency

16. Power to national grid = 70% of generated power

Source: (Southern African Sugar Technologists Association Laboratory Manual, n.d.)(Rajput, 2010)

3.2 Emission analysis

Stoichiometric method

3.3 Financial analysis

Table 3: Financial analysis

Capital structure	40% Owner's equity @ 20%
	60% Loan @ 15%
Cost of 110 bar biomass thermal power plant	NRs. 144 million per MW
Cost of 67 bar biomass thermal power plant	NRs. 120 million per MW
Corporate tax	25%
Salvage value	15%
Life of the project	15 years
Average operating days using bagasse as fuel	120 days
Average operating days using rice husk as fuel	90 days
NEA: per kWh buying cost during dry season	NRs. 8.2
NEA: per kWh selling cost to industry using above 60KV voltage	NRs. 6.25
O & M cost for 120 days of operation	3% of installation cost
O & M cost for 210 days of operation	4% of installation cost

IV. FINDINGS

Replacing conventional low pressure boilers by modern high pressure boilers each sugar mill can contribute surplus energy obtained from bagasse incineration.

4.1 Technical

Calculations were conducted for two different pressure boilers i.e. 67bar and 110 bar boilers. Power contribution to national grid by sugar mill for 120 days is:

Table 4: Energy contribution to national grid by sugar mill, current trend

Industry	Energy Generated (kWh)		Energy to National Grid (kWh)		Power to National Grid (MW)	
	@ 67 bar	@ 110 bar	@ 67 bar	@ 110 bar	@ 67 bar	@ 110 bar
Everest	50978371	65177080	35684860	45623956	12.4	15.9
Shree ram	20850391	26657728	14595273	18660410	5.07	6.48
Indushanker	29714997	37991342	20800498	26593939	7.22	9.23
Eastern	14631539	18706776	10242077	13094743	3.56	4.55
Bagmati	17043083	21789993	11930158	15252995	4.14	5.30
Reliance	12715242	16256745	8900670	11379721	3.09	3.95
Indira	7478577	9561541	5235004	6693079	1.82	2.32
Lumbini	6837556	8741980	4786289	6119386	1.66	2.12
Ganga	7478577	9561541	5235004	6693079	1.82	2.32
Mahakali	8482843	10845520	5937990	7591864	2.06	2.64
Dhankaul	30769001	39338912	21538301	27537239	7.48	9.56
Mahalaxmi	39102273	49993201	27371591	34995241	9.50	12.1
Total	246,082,449	314,622,359	17,225,7714	220,235,652	59.8	76.4

If all the sugar plant are recognized as national power plants (running at their optimum capacity) during dry season

i.e. four months. In addition to bagasse rice husk can be used for 2 months. Power contribution to national grid by sugar mill for 120 days using bagasse and 90 days using rice husk are

Table 5: Energy contribution using bagasse as fuel (for 120 days), full capacity

Industry	Energy Generated (kWh)		Energy to National Grid (kWh)		Power to National Grid (MW)	
	@ 67 bar	@ 110 bar	@ 67 bar	@ 110 bar	@ 67 bar	@ 110 bar
Everest	64102086	61467050	44871460	57369247	16	20
Shree ram	44871460	57369247	31410022	40158473	11	14
Indushanker	57691878	73760460	40384314	51632322	14	18
Eastern	32051043	40978033	22435730	28684623	8	10
Bagmati	12820417	16391213	8974292	11473849	3	4
Reliance	23076751	29504184	16153726	20652929	6	7
Indira	6410209	8195607	4487146	5736925	2	2
Lumbini	14102459	18030335	9871721	12621234	3	4
Ganga	6410209	8195607	4487146	5736925	2	2
Mahakali	12820417	16391213	8974292	11473849	3	4
Dhankaul	38461252	49173640	26922876	34421548	9	12
Mahalaxmi	38461252	49173640	26922876	34421548	9	12
Total	351,279,433	428,630,230	245,895,603	314,383,473	85	109

Table 6: Power contribution using rice husk as fuel (for 90 days)

S.N.	Name of Industry	Power to national Grid(kWh)		Power to Grid (MW)	
		@ 67 bar	@ 110 bar	@ 67 bar	@ 110 bar
1	Everest	48076565	61467050	22	28
2	Shree ram	33653595	43026935	16	20
3	Indushanker	43268908	55320345	20	26
4	Eastern	24038282	30733525	11	14
5	Bagmati	9615313	12293410	4	6
6	Reliance	17307563	22128138	8	10
7	Indira	4807656	6146705	2	3
8	Lumbini	10576844	13522751	5	6
9	Ganga	4807656	6146705	2	3
10	Mahakali	9615313	12293410	4	6
11	Dhankaul	28845939	36880230	13	17
12	Mahalaxmi	28845939	36880230	13	17
	Total	263,459,575	336,839,435	122	156

4.2 Economic Findings

Financial analysis of high pressure boiler operating 210 days using bagasse (for 120 days) and rice husk (for 90 days) as fuel.

4.2.1 110 bar boiler

1. Net present value of the project at the 15% opportunity cost of capital (MARR) will be greater than zero over a period of 15 years of operation. Hence the project appears worth undertaking.
2. MARR 15%
3. Return on total investment will be 41%
4. Return on equity will be 104%.
5. Internal rate of return is 41.22%.
6. Profit margin will be 85%
7. Benefit cost ratio will be 1.98
8. Payback period will be 3 years
9. Break-even point will be 12%

4.2.2 67 bar boiler

1. Net present value of the project at the 15% opportunity cost of capital (MARR) will be greater than zero over a period of 15 years of operation. Hence the project appears worth undertaking.
2. MARR 15%
3. Return on total investment will be 60%
4. Return on equity will be 150%.
5. Internal rate of return is 59.91%.
6. Profit margin will be 88%
7. Benefit cost ratio will be 2.86
8. Payback period will be 2 years
9. Break-even point will be 11.53%

Financial analysis of high pressure boiler operating 120 days using bagasse only as fuel

4.2.3 110 bar boiler

1. Net present value of the project at the 15% opportunity cost of capital (MARR) will be greater than zero over a period of 15 years of operation. Hence the project appears worth undertaking.
2. MARR 15%
3. Return on total investment will be 25%
4. Return on equity will be 63%.
5. Internal rate of return is 24.33%.
6. Profit margin will be 87%
7. Benefit cost ratio will be 1.27
8. Payback period will be 4 years
9. Break-even point will be 3.8%

4.2.4 67 bar boiler

1. Net present value of the project at the 15% opportunity cost of capital (MARR) will be greater than zero over a period of 15 years of operation. Hence the project appears worth undertaking.
2. MARR 15%
3. Return on total investment will be 38%
4. Return on equity will be 96%.
5. Internal rate of return is 38.12%.
6. Profit margin will be 95%
7. Benefit cost ratio will be 1.92
8. Payback period will be 3 years
9. Break-even point will be 2.47%

4.3 Emissions findings:

Table 7: Emissions at full capacity

S.N.	Name of Industry	Rice Husk (MT)	Bagasse (MT)	CO ₂ (MT)	N ₂ O (MT)	SO ₂ (MT)
1	Everest	106239	204000	315798	17814	17406
2	Shree ram	74367	142800	221059	12470	12184
3	Indushankar	95615	183600	284218	16033	15666
4	Eastern	53119	102000	157899	8907	8703
5	Bagmati	21248	40800	63160	3563	3481
6	Reliance	38246	73440	113687	6413	6266
7	Indira	10624	20400	31580	1781	1741
8	Lumbini	23372	44880	69476	3919	3829
9	Ganga	10624	20400	31580	1781	1741
10	Mahakali	21248	40800	63160	3563	3481
11	Dhankaul	63743	122400	189479	10689	10444
12	Mahalaxmi	63743	122400	189479	10689	10444

V. CONCLUSION

Bagasse, otherwise a refuse, if used as cogeneration fuel, is proved to have been technically feasible, economically viable for the competitive industrial environment of sugar industries, environmentally friendly because of greenhouse neutral emissions and acceptable regarding social matters.

ACKNOWLEDGEMENT

The authors gratefully acknowledge the contributions of everyone involved during the planning and organizing the background for this project.

REFERENCES

- [1] Agribusiness Promotion and Statistics Division, 2011-2012. *Statistical Information on Nepalese Agriculture*, Kathmandu: Government of Nepal.
- [2] Baker, R., 1997. *Bagasse Combustion in Sugar Mill*, North Carolina: Environmental Protection Agency.
- [3] Central Bureau of Statistics, 2012. *Statistics information on Nepalese agriculture*.
- [4] Chilime Hydropower Pvt. Ltd., 2012. *Chilime The Pioneer, Annual Report 2011-2012*, Kathmandu: Chilime Hydropower Pvt. Ltd..

- [5] Deepchand, D. K., 2001. *Bagasse-Based Cogeneration in Mauritius*, Nairobi: African Energy Policy Research Network.
- [6] International Rice Research Institute, 2009. *Rice Knowledge Bank*. [Online]
Available at: <http://www.knowledgebank.irri.org/>
[Accessed 12 April 2013].
- [7] Kolar Seenappa Venkatesh, A. S. R., 2010. Development and Installation of High Pressure Boilers for Co-Generation Plant in Sugar Industries. *Smart Grid and Renewable Energy*, Volume 1, pp. 51-53.
- [8] Mitra Phol Bio Power, 2012. *Biomass Power: High Efficiency Boiler Technology for Sugar Industry*. Thailand, MITRA PHOL GROUP.
- [9] Nepal Electricity Authority, 2013. *A Year in Review- Fiscal 2012/2013*, Kathmandu: NEA.
- [10] NIFES, 2011. *Cogeneration training material*. [Online].
- [11] Olivério, J. & F. F. e. a., 2010. Cogeneration – A New Source of Income for Sugar Mill. *Proc. Int. Soc. Sugar Cane Technol.*, Volume 27.
- [12] Osler, C., 2011. *Diesel & Thermal Electricity Generation Options*, Winnipeg: InterGroup Consultants Ltd..
- [13] Park, C. S., 2012. *Contemporary Engineering Economics*. 5th ed. New Jersey: Pearson Education Inc..
- [14] Rajput, R. K., 2010. *Thermal Engineering*. 7th ed. New Delhi: Laxmi Publication.
- [15] Southern African Sugar Technologists Association Laboratory Manual, n.d. *Sugar Engineers*. [Online]
Available at: <http://www.sugartech.co.za/>
- [16] United States Environment Protection Agency, 2010. *Overview of greenhouse gases*. [Online]
Available at: <http://epa.gov/climatechange/ghgemissions/gases/ch4.html>
[Accessed 2013].
- [17] World Alliance for Decentralized Energy, 2004. *Bagasse cogeneration - global review and potential*.

BIOGRAPHIES

Mr. Sajjan Pokhrel is a graduate of Mechanical Engineering from IOE, Pulchowk. He worked with AEPC as a research consultant. He is currently Assistant Lecturer at Kantipur City College.

Mr. Saroj Bhattarai is a graduate of Mechanical Engineering from IOE, Pulchowk. He worked with AEPC as a research consultant.

Mr. Srijan Rajbanshi is a graduate of Mechanical Engineering from IOE, Pulchowk. He is currently working at E and T, Nepal.

Mr. Yogesh Bhusal is a graduate of Mechanical Engineering from IOE, Pulchowk. He is currently working as program coordinator of civil department at Universal Science College.